

EST Project Statements (UCS321)

1. Honeywell aims to design a neural network-based predictive maintenance model for smart factories. Develop a robust model to predict potential equipment failures in advance using multi-sensor heterogeneous data streams, incorporating anomaly detection, feature fusion, and real-time inference, so that maintenance can be scheduled optimally while minimizing false alarms and improving overall system efficiency.
2. ABB Power Systems seeks to improve the reliability of high-voltage transmission lines by detecting faults in real time. Using noisy and imbalanced time-series voltage and current waveform data, develop an optimized 1D Convolutional Neural Network (CNN) model with attention mechanisms to accurately classify different types of faults such as short-circuit, overload, and ground fault under varying operating conditions.
3. Tata Steel Ltd. is automating quality inspection in its manufacturing process. Develop a high-performance CNN-based image recognition system capable of detecting micro-level surface cracks in steel plates from high-resolution images, incorporating data augmentation, transfer learning, and explainable AI techniques to ensure robustness and interpretability in real-time production environments.
4. Biocon Ltd. is working on automated diagnosis of blood samples. Develop an advanced CNN model to classify red blood cell images as healthy or malaria-infected, addressing class imbalance, image noise, and variability, while incorporating model interpretability (e.g., Grad-CAM) to assist medical experts in validating predictions.
5. Indian Oil Corporation seeks to forecast the Air Quality Index (AQI) around its refinery zones. Using incomplete and multivariate historical pollutant data, develop a hybrid ARIMA and LSTM-based forecasting model with feature selection, seasonal decomposition, and hyperparameter tuning to accurately predict AQI values and support proactive emission control strategies.
6. BSES Rajdhani Power Ltd. requires accurate short-term electricity demand forecasting to ensure efficient grid management. Develop a scalable LSTM-based forecasting system using large-scale temporal datasets, integrating weather variability, anomaly handling, and multi-step forecasting to improve prediction accuracy and grid stability.
7. Caterpillar Inc. is enhancing predictive maintenance for health monitoring of heavy machinery such as turbines and excavators. Using high-frequency vibration sensor data, develop a deep learning model (CNN/LSTM) that performs feature extraction, fault classification, and early-stage fault detection under noisy industrial conditions, ensuring high reliability and low latency.
8. Bosch Engineering and Business Solutions is developing gesture-based human-machine interaction systems for factory automation. Using accelerometer and gyroscope readings from wearable sensors, develop a CNN-LSTM model that can handle sequential dependencies, sensor noise, and real-time constraints to accurately classify complex human activities and transitions between them.
9. Reliance Chemicals Ltd. seeks to optimize chemical production by predicting reaction yields. Using nonlinear and high-dimensional physicochemical reaction data (temperature, pressure, reactant concentrations), develop a neural network regression model with feature engineering, sensitivity analysis, and optimization capabilities to accurately estimate chemical reaction yield and improve process efficiency.
10. Rolls-Royce Aerospace is focused on predictive maintenance of aircraft engines. Develop a deep learning model (RNN/LSTM) to predict the Remaining Useful Life (RUL) of turbofan engines using multi-sensor time-series data, incorporating sequence modeling, degradation pattern learning, uncertainty estimation, and domain adaptation to ensure accurate and reliable maintenance scheduling.

- Dataset can be obtained from standard platforms (e.g., Kaggle, GitHub).
- Data pre-processing can be applied as per project scope and relevance.
- Performance evaluation parameters can be chosen based on problem type.
- A flow diagram must be given with Pre-processing and visualization steps clearly documented.